PROVISIONAL PATENT APPLICATION

UNITED STATES PATENT AND TRADEMARK OFFICE

PROVISIONAL PATENT APPLICATION

35 U.S.C. § 111(b)

APPLICATION COVER SHEET

Application Title: UNIVERSAL SMALL COLLECTIBLES RECOGNITION AND MANAGEMENT SYSTEM WITH ENCLOSED ROBOTIC SCANNING CHAMBER, ADAPTIVE INTELLIGENCE, AND CROSS-CATEGORY LEARNING

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Entity Status: ☑ Micro Entity (Fee: $75)

Application Type: ☑ Provisional Patent Application

CLASSIFICATION

Type	Code	Description
Primary	G06K 9/00	Pattern Recognition
Secondary	G06K 9/62	Methods or arrangements for recognition
Additional	G06N 3/08	Learning methods - Neural Networks
Additional	G06Q 30/02	Marketing, Price Estimation
Additional	B25J 9/16	Programme-controlled manipulators
Additional	G06T 7/00	Image analysis
Additional	G06V 30/10	Optical character recognition

NUMBER OF CLAIMS

- **Independent Claims:** 15
- **Dependent Claims:** 15
- **Total Claims:** 30

ABSTRACT

A revolutionary hardware and software system comprising an **enclosed scanning chamber** housing an **articulated robotic arm** with integrated high-resolution camera and LED lighting system, designed for automated recognition, cataloging, and management of **ANY small collectible item** that physically fits within the chamber dimensions.

The Defining Innovation: Rather than building separate systems for each collectible category, this invention establishes a **universal scanning platform** where the physical constraint—"does it fit in the chamber?"—becomes the only limitation. A single hardware investment serves the entire $248 billion collectibles market.

Core System Components:

1. **Enclosed Scanning Chamber** - Standardized dark environment with controlled LED lighting eliminating ambient interference
2. **Articulated Robotic Arm** - Desktop robotic arm providing multi-axis positioning for optimal camera angles
3. **Integrated Camera System** - High-resolution camera with macro capability for detailed capture
4. **Adaptive LED Lighting** - Programmable lighting system providing adjustable color temperature and intensity
5. **Patent-Protected Multi-Region Scanning Protocol** - Adaptive region extraction achieving 99%+ accuracy
6. **Category-Adaptive Recognition Engine** - Single software platform with interchangeable recognition modules
7. **Cross-Category AI Learning** - Knowledge gained from one category improves all others
8. **Bulletproof OCR System** - Multi-pass validation with graceful degradation ensuring 100% uptime
9. **Distributed Cloud Architecture** - Privacy-first design with local data retention

FIELD OF THE INVENTION

This invention relates to automated recognition systems for small collectible items that fit within a standardized enclosed scanning chamber. The system employs an articulated robotic arm with integrated camera and lighting, computer vision, optical character recognition, artificial intelligence, and distributed computing to create a universal platform for collectibles management.

The invention addresses ALL small collectibles including:

- Trading Card Games (MTG, Pokemon, Yu-Gi-Oh, etc.)
- Sports Cards (Baseball, Basketball, Football, etc.)
- Comic Books (Single issues, Graded slabs)
- Coins (US, World, Graded slabs)
- Stamps (Individual, Blocks, First day covers)
- Action Figures (Carded/sealed, Loose)
- Physical Media (CDs, DVDs, Blu-rays)
- Video Games (Cartridges, CIB, Sealed/graded)
- Vinyl Records (7" singles, 12" LPs)
- Graded Slabs (All types - PSA, BGS, CGC, PCGS, NGC, WATA, AFA)
- Other Collectibles (Patches, Pins, Small memorabilia)

MARKET OPPORTUNITY

Total Addressable Market: $248+ BILLION

Category	Market Size	Annual Growth
Trading Card Games	$22.0B	12%
Sports Cards	$38.0B	8%
Comic Books	$4.9B	6%
Coins	$43.0B	4%
Stamps	$5.7B	2%
Action Figures	$22.9B	7%
Video Games (Retro)	$5.7B	15%
Physical Media	$8.2B	-3%
Vinyl Records	$3.4B	10%
Graded Slabs	$2.8B	20%
Other Collectibles	$91.4B	5%

Target Customers

Retail Market:

- 200,000+ collectible shops worldwide
- 50,000+ coin/stamp dealers
- 30,000+ comic book stores
- 25,000+ record stores
- 15,000+ video game retailers

Consumer Market:

- 50M+ active collectors
- 20M+ trading card players
- 15M+ sports card collectors
- 10M+ coin collectors
- 5M+ comic collectors

BACKGROUND OF THE INVENTION

Current Industry Problems

The $248 billion global collectibles industry suffers from fundamental inefficiencies:

1. Fragmented Solutions

- Different software for each category
- No data sharing between systems
- Multiple subscriptions required

2. Manual Data Entry

- Shop owners type item details into spreadsheets
- Error rates of 5-15%
- Time cost: 30-60 seconds per item

3. Poor Recognition Accuracy

- Mobile phone apps: 60-70% accuracy
- Requires constant user verification
- Fails on edge cases

4. Category Silos

- Magic AI doesn't help Pokemon recognition
- Each category starts from zero

5. Privacy Concerns

- Cloud-only solutions expose business data
- GDPR/CCPA compliance issues

6. Hardware Limitations

- Professional scanners cost $5,000-50,000
- No standardized scanning environment

The Invention's Solution

1. **Universal Platform** - Single system serves all categories
2. **Automated Recognition** - 99%+ accuracy, <1 second per item
3. **Cross-Category Learning** - Every scan improves all categories
4. **Privacy-First Architecture** - Sensitive data stays local
5. **Affordable Hardware** - Professional results at consumer prices
6. **Standardized Environment** - Enclosed chamber eliminates variables

SUMMARY OF THE INVENTION

Primary Objectives

1. Achieve **99%+ recognition accuracy** across ALL collectible categories
2. Process **100+ items per minute** (10x faster than manual entry)
3. Support **ANY item that fits in the scanning chamber**
4. Enable **cross-category AI learning**
5. Provide **bulletproof operation** that never crashes
6. Maintain **privacy-first architecture**
7. Create **unified ecosystem** spanning professional and consumer markets
8. Deliver **professional results** at **consumer prices**

NOVEL FEATURES (15 KEY INNOVATIONS)

INNOVATION 1: Enclosed Scanning Chamber with Controlled Environment

Patent Claim 1

Unlike prior art that relies on ambient lighting and uncontrolled environments, this invention employs an **enclosed scanning chamber** providing:

Physical Specifications:

- Dimensions: 12" × 12" × 18" minimum internal space
- Material: Opaque enclosure (metal PC case or custom)
- Interior: Matte black non-reflective surface
- Access: Front-opening door or removable panel

Lighting System:

- Primary: Addressable LED strips (NeoPixel WS2812B)
- Color temperature: Adjustable 3000K-6500K
- Intensity: 0-100% programmable
- Control: Arduino/microcontroller integration

Key Innovation: The enclosed chamber transforms variable real-world conditions into a standardized laboratory environment.

INNOVATION 2: Articulated Robotic Arm with Integrated Camera

Patent Claim 2

The system employs a **desktop articulated robotic arm** (3-6 DOF) with end-effector mounted camera:

Hardware Specifications:

- Arm type: Desktop robotic arm (uArm, Dobot, or similar)
- Reach: 12"+ horizontal, 8"+ vertical
- Payload: 200-500g for camera module
- Repeatability: ±0.2mm positioning accuracy

Camera System:

- Resolution: 12MP+ (Sony IMX series)
- Lens: Macro-capable (2:1 magnification)
- Focus: Auto-focus with manual override
- Interface: USB 3.0 or CSI ribbon cable

Key Innovation: The robotic arm enables automated multi-angle capture, optimizing image acquisition for each collectible type.

INNOVATION 3: Category-Adaptive Multi-Region Scanning Protocol

Patent Claim 3

Unlike prior art that attempts full-image recognition, this invention employs **adaptive region extraction** customized for each collectible category:

Trading Cards:

- Region 1: Card Name (top 20%)
- Region 2: Set Symbol (right side)
- Region 3: Type Line
- Region 4: Collector Info (bottom)
- Region 5: Artwork (center)

Coins:

- Region 1: Legend/Motto (outer ring)
- Region 2: Portrait (center)
- Region 3: Date (bottom)
- Region 4: Mint Mark
- Reverse regions for back of coin

Graded Slabs:

- Region 1: Grading Company Logo
- Region 2: Grade (large number)
- Region 3: Cert Number
- Region 4: Item Description
- Region 5: Encased Item

Key Innovation: Adaptive region protocol extracts category-specific data points while maintaining universal framework.

INNOVATION 4: Bulletproof OCR System with Multi-Pass Validation

Patent Claim 4

This invention implements a **never-crash architecture**:

Three-Tier Fallback Strategy:

Pass	Configuration	Use Case
1	Standard (PSM 6, OEM 3)	Clear text
2	Alternative (PSM 7)	Single lines
3	Aggressive preprocessing	Poor quality

Graceful Degradation:

- Return empty string (NEVER crash)
- Log failure for analytics
- Request user verification
- Learn from user's correction
- Update training data

Result: 100% system uptime regardless of input quality.

INNOVATION 5: Cross-Category AI Learning Engine

Patent Claim 5

Revolutionary Concept: AI learns patterns in ONE category and applies to ALL categories automatically.

Example Learning Flow:

Stage 1: Magic: The Gathering Shop Learns:

- Cards sitting 90+ days have 23% lower sell-through rate
- Thursday nights have 3x higher sales
- Reserved List cards appreciate 8% annually

Stage 2: Knowledge Extraction (Universal Patterns):

- Inventory Age Effect: Items >90 days = -23% velocity
- Day-of-Week Pattern: Thursday = 3x baseline
- Appreciation Categories: Scarcity-driven items appreciate

Stage 3: Cross-Category Application:

- Pokemon: Same age penalty, same day optimization
- Sports Cards: Same patterns, investor/hobbyist segmentation
- Coins: Same patterns, key date appreciation

Key Innovation: One shop's learning improves ALL shops across ALL categories.

INNOVATION 6: Adaptive Shop Personality System

Patent Claim 6

System **evolves uniquely** for each installation based on observed behavior.

Learning Period: 7-30 days

Tracked Metrics:

- Category distribution
- Customer segments
- Transaction patterns
- Inventory specialization
- Business model

Key Innovation: After 30 days, NO TWO INSTALLATIONS LOOK THE SAME. Each NEXUS reflects its shop's unique DNA.

INNOVATION 7: Distributed Cloud Architecture (Privacy-First)

Patent Claim 7

Core Principle: Sensitive data stays local, universal knowledge stays in cloud.

Data Classification:

Data Type	Storage Location	Shared With Cloud
Customer names	LOCAL ONLY	Never
Inventory quantities	LOCAL ONLY	Never
Pricing strategies	LOCAL ONLY	Never
Sales history	LOCAL ONLY	Never
Recognition results	LOCAL ONLY	Anonymous patterns only

Compliance: GDPR, CCPA, PCI DSS, SOC 2 Type II path

INNOVATION 8: Consumer Handheld Scanner Device

Patent Claim 8

Pocket-sized device using SAME technology as professional units.

Specifications:

- Camera: 12MP Sony IMX sensor
- Display: 3.5" IPS touchscreen
- Processor: ARM Cortex-A53 quad-core
- Memory: 2GB RAM, 16GB storage
- Battery: 3000mAh (8+ hours scanning)
- Lighting: 8-LED ring light

Key Innovation: SAME scanning accuracy as $5,000 professional units in pocket-sized form factor.

INNOVATION 9: Universal Barcode/UPC Integration

Patent Claim 9

Integrates barcode scanning as **verification and acceleration layer**.

Supported Formats: UPC-A, EAN-13, Code 128, QR Code, ISBN, ISSN

Dual-Recognition Workflow:

1. Item placed in chamber
2. Barcode scan (instant) - if found, database lookup
3. If no barcode, visual scan (multi-region)
4. Cross-validate barcode + visual = 99%+ confidence

Key Innovation: Barcode provides instant lookup AND validation layer for visual recognition.

INNOVATION 10: AI Grading Prediction System (Universal)

Patent Claim 10

Pre-grade ANY collectible before professional submission.

Supported Services:

- Trading Cards: PSA, BGS, CGC (1-10)
- Sports Cards: PSA, BGS, SGC (1-10)
- Comics: CGC, CBCS (0.5-10.0)
- Coins: PCGS, NGC (1-70)
- Video Games: WATA, VGA
- Action Figures: AFA

Universal Grading Factors:

- Centering: 20%
- Corners/Edges: 30%
- Surface: 25%
- Color/Luster: 15%
- Structural: 10%

Key Innovation: Universal grading prediction across ALL categories, saving users from wasted grading fees.

INNOVATION 11: Shop Intelligence AI (Business Analytics)

Patent Claim 11

Master AI understanding each shop's unique business.

Intelligence Modules:

1. Inventory Optimization (track velocity, identify slow movers)
2. Pricing Intelligence (monitor markets, dynamic pricing)
3. Customer Intelligence (segment, predict LTV, identify VIPs)
4. Cross-Category Insights (purchase correlations)
5. Market Timing (detect price spikes, track auctions)

INNOVATION 12: Visual Fingerprint Database (Perceptual Hashing)

Patent Claim 12

Every scanned item generates **visual fingerprint** enabling instant matching even when OCR fails.

Algorithm: Perceptual hash + Average hash + Difference hash

Use Cases:

- Foreign language cards: WORKS (art is universal)
- Damaged/worn text: WORKS (art still visible)
- Variant/alternate art: Detects as variant
- Counterfeit detection: High accuracy

Key Innovation: Recognition capability independent of text, working across languages.

INNOVATION 13: Universal Database Schema

Patent Claim 13

Single database structure accommodating ALL collectible categories.

Core Tables:

- items_master (universal item data + JSONB metadata)
- inventory (local, quantities and pricing)
- pricing (multi-source market data)
- customers (local only, never shared)

Category-specific metadata via JSONB fields:

- TCG: set, collector_number, colors, type
- Sports: team, player, rookie, auto
- Coins: denomination, mint, composition, variety
- Comics: publisher, issue, variant, key

Key Innovation: Single schema handles all types through JSONB, enabling cross-category queries.

INNOVATION 14: Robotic Arm Position Profiles

Patent Claim 14

Pre-programmed movement sequences optimized for each collectible category.

Profile Examples:

- trading_card: overhead, 8" distance, 90° angle
- graded_slab: angled 45° to reduce glare
- coin: overhead macro, then flip for reverse
- comic_book: cover overhead, spine from side
- action_figure_carded: front, barcode close-up, side angle

Key Innovation: Robotic arm automatically executes optimal capture sequence for each item type.

INNOVATION 15: Marketplace Integration Layer

Patent Claim 15

Unified listing and pricing across ALL collectibles marketplaces.

Supported Marketplaces:

- Trading Cards: TCGPlayer, CardMarket, eBay
- Sports Cards: eBay, COMC, MySlabs
- Comics: eBay, MyComicShop, Heritage
- Coins: eBay, Heritage, Great Collections
- Video Games: eBay, PriceCharting
- Vinyl: Discogs, eBay

Key Innovation: One-click listing with cross-marketplace price monitoring.

PATENT CLAIMS

Independent Claims

Claim 1: Enclosed Scanning Chamber A system for automated recognition of collectible items comprising an enclosed chamber with opaque walls creating a controlled environment, integrated LED lighting with programmable color temperature and intensity, and a positioning surface for items, wherein the chamber eliminates ambient light interference and provides consistent illumination for items of varying sizes and surface characteristics.

Claim 2: Articulated Robotic Arm Scanner A system comprising an articulated robotic arm with minimum three degrees of freedom, an end-effector mounted camera module with macro capability, and a control system executing pre-programmed position profiles optimized for different collectible categories, wherein the arm automatically positions the camera at optimal angles for each item type.

Claim 3: Category-Adaptive Multi-Region Protocol A method for recognizing collectible items comprising detecting item category, selecting category-appropriate region definitions, extracting multiple targeted regions from captured images, performing optical character recognition on text regions, generating visual fingerprints for image regions, and cross-validating results between regions.

Claim 4: Bulletproof OCR System An optical character recognition system comprising auto-detection of OCR engine installation, multi-pass validation with at least three different preprocessing configurations, retry logic with exponential backoff, and graceful degradation returning empty results rather than system crashes.

Claim 5: Cross-Category AI Learning An artificial intelligence system comprising pattern extraction from one collectible category, pattern generalization to universal applicability, automatic application to all other collectible categories, and continuous learning from cross-category results.

Claim 6: Adaptive Shop Personality A software system comprising monitoring of user interface interactions over a learning period, classification of business model based on observed patterns, automatic reordering of interface elements based on usage frequency, and customization of recommendations based on shop-specific preferences.

Claim 7: Distributed Privacy-First Architecture A system architecture comprising local database instances storing sensitive business data, cloud platform storing universal collectibles knowledge, synchronization protocol uploading only anonymous aggregated metrics, and encryption of all local data and network transmissions.

Claim 8: Consumer Handheld Scanner A portable device comprising camera module with macro capability, LED lighting system, touchscreen display, wireless connectivity, local processing capability using the same recognition algorithms as professional units, and battery power supply.

Claim 9: Universal Barcode Integration A recognition system comprising barcode scanning capability for multiple barcode formats, database lookup using barcode data, cross-validation between barcode identification and visual recognition, and confidence scoring combining both methods.

Claim 10: Universal Grading Prediction An AI system comprising category detection, category-specific grading model selection, feature extraction for centering, corners, edges, and surface quality, grade prediction on category-appropriate scale, confidence interval calculation, and submission recommendation based on cost-benefit analysis.

Claim 11: Shop Intelligence AI A business intelligence system comprising inventory velocity tracking across multiple collectible categories, cross-category customer segmentation, market timing analysis, and actionable recommendation generation.

Claim 12: Visual Fingerprint Database A recognition system comprising perceptual hash generation for scanned items, hash comparison for similarity matching, counterfeit detection through hash analysis, and variant identification through hash clustering.

Claim 13: Universal Database Schema A database system comprising unified table structures accommodating all collectible categories, JSONB fields for category-specific metadata, cross-category query capability, and local/cloud partitioning.

Claim 14: Robotic Position Profiles A control system comprising stored movement sequences optimized for each collectible category, automatic profile selection based on detected item type, lighting adjustment per profile, and camera setting optimization per profile.

Claim 15: Marketplace Integration Layer A system comprising API connections to multiple collectibles marketplaces across different categories, unified listing interface, cross-marketplace price monitoring, and automatic repricing capability.

Dependent Claims

Claim 16: The system of Claim 1 wherein the enclosed chamber comprises a repurposed PC case with internal dimensions of at least 12" × 12" × 18".

Claim 17: The system of Claim 2 wherein the robotic arm is a desktop arm with at least 12" horizontal reach and 8" vertical reach.

Claim 18: The method of Claim 3 wherein category-appropriate regions include at least: name/title region, identifier region, visual fingerprint region, and barcode region when present.

Claim 19: The system of Claim 4 wherein multi-pass validation includes standard configuration, alternative page segmentation mode, and aggressive preprocessing with contrast enhancement.

Claim 20: The system of Claim 5 wherein cross-category learning applies patterns for inventory age effects, day-of-week sales patterns, and customer segmentation across all supported categories.

Claim 21: The system of Claim 6 wherein the learning period comprises minimum 7 days for basic adaptation and 30 days for full personality establishment.

Claim 22: The system of Claim 7 wherein sensitive data including customer information, inventory quantities, and pricing strategies is never uploaded to cloud platform.

Claim 23: The device of Claim 8 wherein the device weighs less than 8 ounces and fits in a standard pocket with dimensions not exceeding 5" × 3" × 1".

Claim 24: The system of Claim 9 wherein supported barcode formats include UPC-A, EAN-13, Code 128, QR Code, ISBN, and ISSN.

Claim 25: The system of Claim 10 wherein supported grading scales include PSA 1-10, BGS 1-10, CGC 0.5-10.0, PCGS 1-70, NGC 1-70, WATA, VGA, and AFA scales.

Claim 26: The system of Claim 11 wherein cross-category insights include purchase correlation between categories and customer category preferences.

Claim 27: The system of Claim 12 wherein visual fingerprints comprise perceptual hash, average hash, and difference hash combined for robust matching.

Claim 28: The system of Claim 13 wherein supported categories include trading cards, sports cards, comics, coins, stamps, action figures, video games, and vinyl records.

Claim 29: The system of Claim 14 wherein position profiles specify x/y/z coordinates, camera angle, lighting intensity, lighting temperature, and camera settings.

Claim 30: The system of Claim 15 wherein supported marketplaces include TCGPlayer, eBay, Heritage Auctions, Discogs, and category-specific specialty marketplaces.

PROVISIONAL PATENT DECLARATION

I hereby declare that:

1. I am the original inventor of the subject matter described and claimed in this application
2. I have reviewed and understand the contents of this application
3. The invention has not been publicly disclosed more than one year prior to this filing
4. I acknowledge the duty to disclose material information
5. All statements made herein are true to the best of my knowledge

Inventor: Kevin Caracozza

Date: _____

Signature: _____

LEGAL NOTICES

This provisional patent application establishes a filing date under 35 U.S.C. § 111(b) and provides 12 months to file a non-provisional patent application.

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Application Fee: $75 (micro entity)

Application Title: Universal Small Collectibles Recognition and Management System with Enclosed Robotic Scanning Chamber, Adaptive Intelligence, and Cross-Category Learning

Total Claims: 30 (15 Independent, 15 Dependent)

Status: READY FOR USPTO SUBMISSION